

Golf Cart Progress

CAN, Networking and Localization

NEWSLab NTU · August 2026

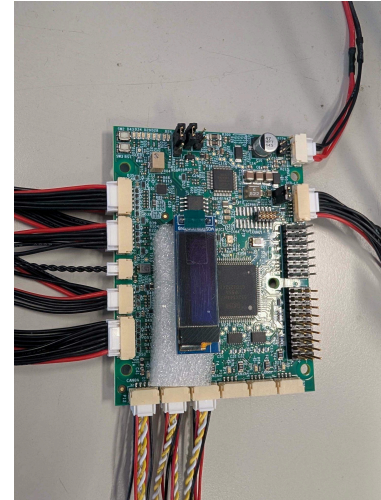
Overview

1. **Multi-host setup** — splitting the stack across Advantech and AGX Orin
2. **TSN** — target architecture and what each machine can support
3. **IMU CAN bus** — wiring failure and the remake
4. **Localization** — NTU campus logging simulation and NDT tuning

First three sections continue the 2026-08-05 report. Localization is new work since then.

Multi-host setup

Multi-host Setup (WIP)



NXP CANHUBK344
Safety Island

The rationale behind the multi-host setup

Constrained computation power

The Advantech drives three mono oToBrite cameras and two LiDARs. It burns. The ZED X stereo + depth camera moves to the AGX Orin.

Driver conflict

The oToBrite and ZED camera drivers cannot be installed together on one machine — they share the same kernel module name.

Hardware limitation

The ZED camera requires a ZEDLink capture card, which only works on Orin.

Challenges

Lack of multi-host support in ROS 2

ROS 2 launch files are designed for a single machine. ROS 1 supported machine tags in launch files; ROS 2 dropped it, moving the multi-host responsibility to the user side.

Non-standard RMW setup

Autoware's recommended Cyclone DDS and Zenoh settings no longer fit. Node and topic discovery has to work across both machines.

Orchestration work is needed

Advantech and Orin run disjoint sets of nodes that together make up Autoware. Start and stop must be synchronous and singleton, with no orphans left after termination.

Multi-host launch method

Add a host argument to launch files

`host:=...` switches each node on or off per machine.

Employ systemd services

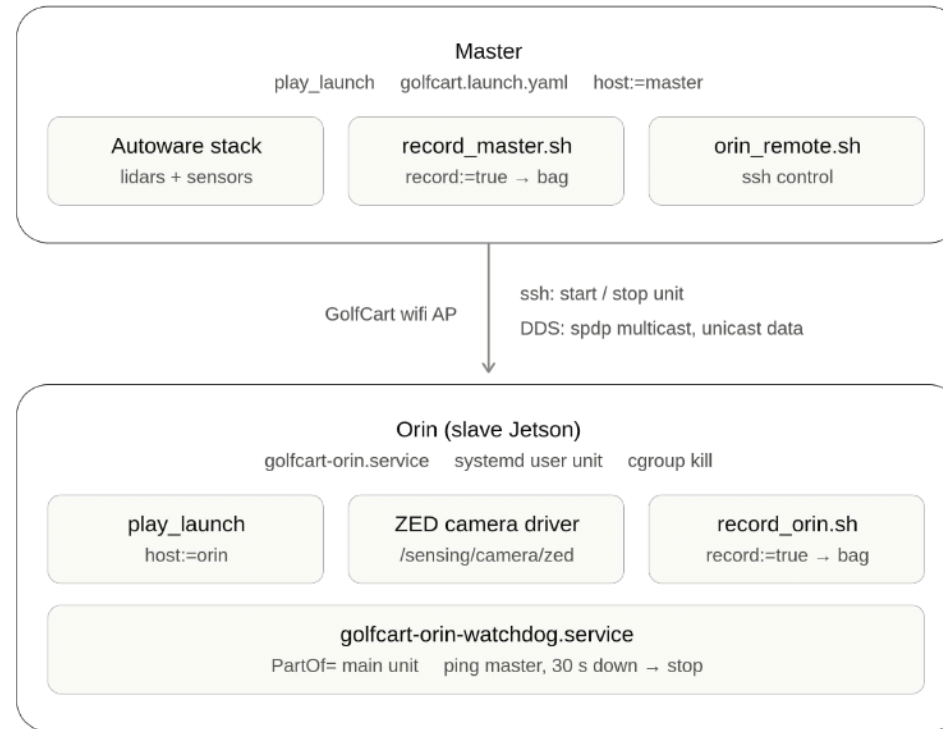
The launch command is wrapped in a systemd user unit, `golfcart-orin.service`. systemd gives singleton startup, clean termination and full logs.

Orphan prevention using cgroups

Spawned nodes stay in a control group jail — nothing escapes it, and `systemctl stop` captures them all.

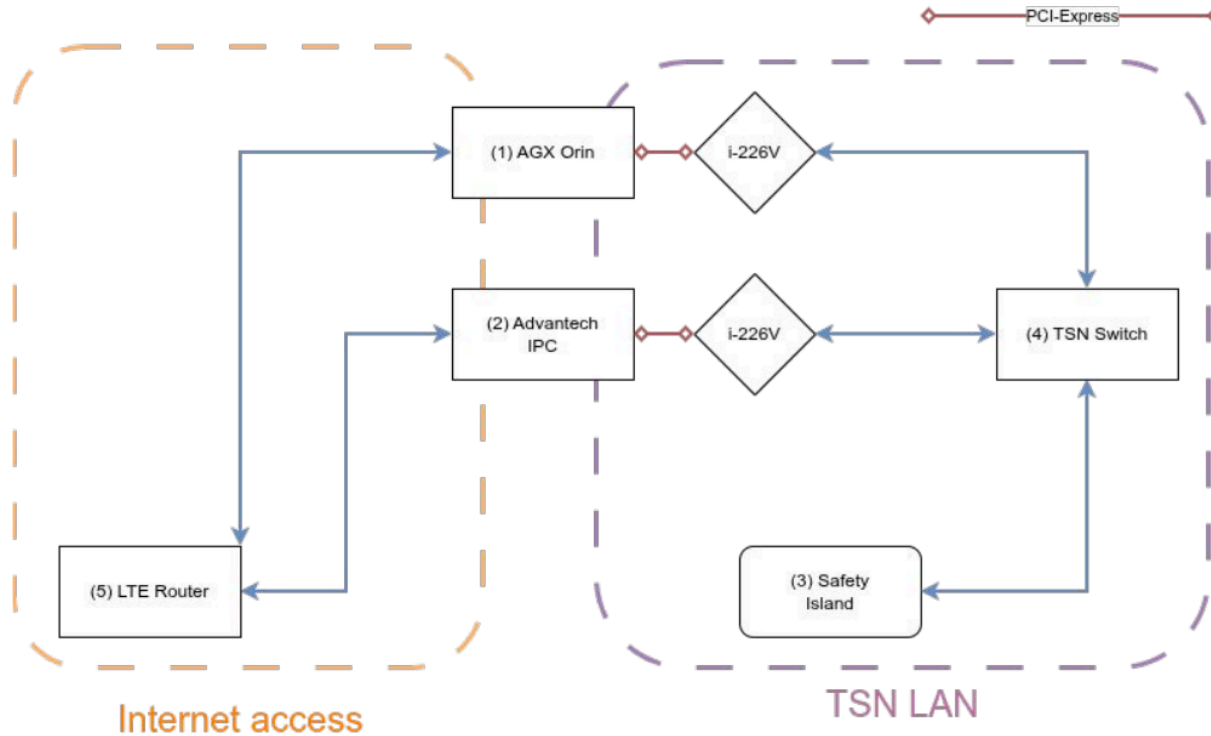
[Since the draft](#): a watchdog unit on the Orin stops the stack if the master disappears, and `just service-install` provisions both hosts.

Multi-host launch



TSN setup

Target Time-Sensitive Network architecture



Design options towards TSN

All hosts in the TSN must have a network card with TSN features:

- **gPTP** — generalized precision time protocol
- **802.1Qbv** — enhancements for scheduled traffic
- **802.1Qav** — credit-based shaper

A TSN switch is necessary because there are three machines in the network.

The switch cannot be replaced by a normal hub, due to PTP requirements.

TSN support on the machines we have

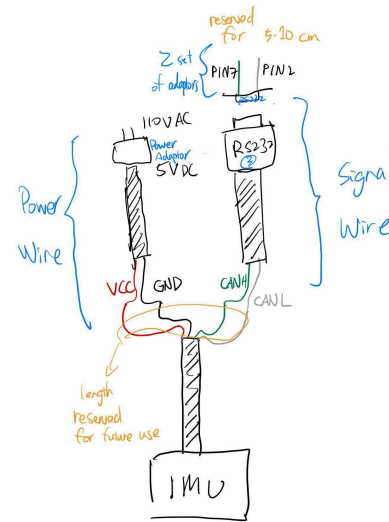
AGX Orin	I226 NIC attached on the PCIe slot
Advantech	NIC does not support TSN. PCIe is not available and may require slot extension.
NXP safety island	Ethernet T1 socket requires a converter to standard Ethernet TX.
AGX Thor	Supports built-in TSN, but has no PCIe slot. A candidate to replace the AGX Orin.

IMU CAN bus fixing

The CAN bus wiring issue



Broken CAN bus connector



The CAN wire remake

CAN wiring remake

Rationale

The IMU's CAN bus connector had exposed resistors and connection points, which were damaged while tidying up the chassis wiring.

Remake

A manufacturer will redo the entire wiring, replacing it with plastic-coated, less easily damaged cable.

Connector change

The wiring moves to an RS232 female connector, with a fabricated RS232 male connector on the host side.

Localization: NDT on the NTU map

NTU campus logging simulation

Two recorded runs per session — Advantech and Orin — merged into one bag, and replayed against a merged r01 + r02 point cloud map.

Sets CSIE-1, CSIE-2, BLVD-1

Map 6.6 M points, MGRS 51RUH, r01 + r02 merged

LiDAR Velodyne VLP-32C, primary for NDT

IMU ZED X built-in — the XSens was not working during the runs

GNSS switched off; initialization is a saved pose

One command runs it end to end: `just ntu-test run CSIE-1` brings up the bag, the stack, RViz and the initial pose in the required order.

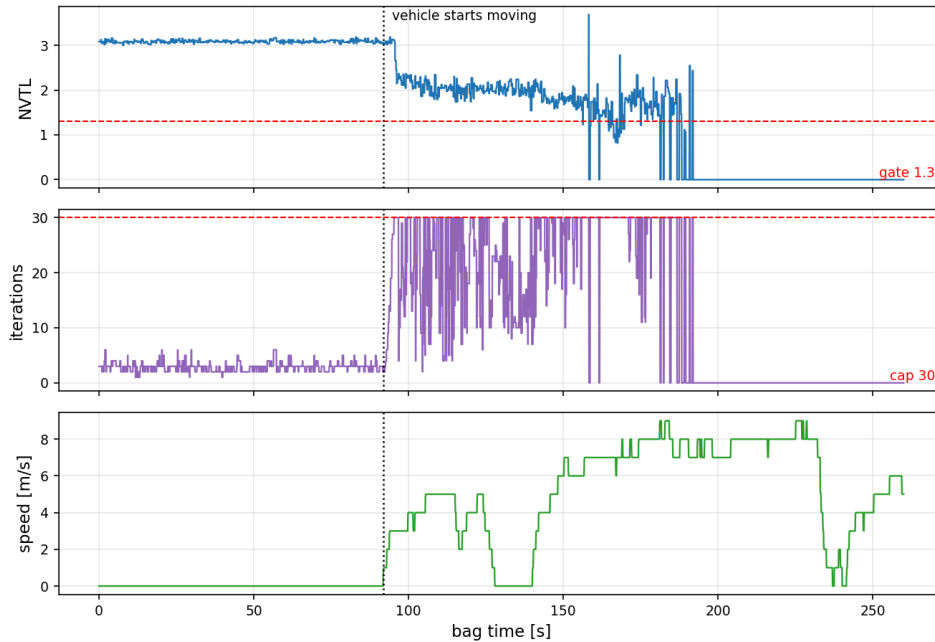
NDT tuning: what the crop range was costing

Ranked on **scan-to-map residual** — the distance from each live scan point to the nearest map point — not on NDT's own score.

crop	voxel	gate	p50	p95	> 3 m
±20 m	0.5	2.3	0.210 m	1.661 m	0.0%
±60 m	0.5	1.3	0.144 m	0.469 m	0.0%
±60 m	3.0	2.3	0.682 m	2.590 m	12.1%

The ±20 m crop was inherited from AutoSDV and clipped real structure — our returns reach 29 m. Restoring Autoware's ±60 m cut p95 by 72%. The convergence threshold had to move with it: NVTL is a mean per-point score, so it fell while accuracy **improved**.

NDT internals: where tracking breaks



Convergence while parked is good — 0.14 m residual.

Degradation starts **3.5 s after the vehicle first moves**, not at the turn. Iterations hit their cap and the correction NDT applies per scan grows from 0.09 m to about 1 m.

NVTL crosses its gate 50 s later, so the score is a lagging indicator — iteration count is the earlier warning.

Root cause: the recordings, not the tuning

45% of scans are partial rotations

Measured over 300 scans: 55% cover 357°, but 45% cover only 162° and span 2.1 ms. Which half of the world they cover changes frame to frame, so the geometry constraining the fit rotates with it.

Scans arrive 376 ms stale

The EKF stamps its poses at the current time; each cloud reaches NDT 376 ms after its own timestamp — 1.9 m of travel at 5 m/s. Autoware reports this as “Couldn’t interpolate pose”, which is what drives the degraded state in RViz.

Neither is fixable in these bags

The recordings contain decoded clouds only — no raw Velodyne packets — so they cannot be re-decoded with a corrected driver configuration.

Status and next steps

Multi-host	working	systemd units on both hosts, cgroup cleanup, watchdog
TSN	design	Orin ready via I226; Advantech NIC is the blocker
IMU CAN	in progress	wiring remake ordered, RS232 connector change
NDT	partial	converges and tracks; degrades from input defects

Next

- Record raw Velodyne packets, so bags can be re-decoded after a driver fix
- Check the VLP-32C rotation configuration against the hardware
- Locate where the 376 ms scan latency accrues

Q&A